MK-1: an exactly simulated two-qubit ZZ-feature-map fidelity kernel, with machine-checked kernel properties

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Abstract

MK-1 is a small, self-contained implementation of a quantum fidelity kernel on two qubits: a ZZ-type feature map $U(x)$, the compute–uncompute estimator $K(x,x') = |\langle 00|U(x')^\dagger U(x)|00\rangle|^2$, and a visualisation of the kernel-SVM pipeline. The construction is not new; it follows Havlíček et al. [1] and Schuld–Killoran [2]. This note records exactly what the code computes, documents a defect in the original script (its "entangling" gate was a no-op, so the kernel factorised), and lists what is machine-checked in Lean 4 with Mathlib: for any unitary feature map, $K(x,x) = 1$, symmetry, $0 \le K \le 1$, compute–uncompute equals state fidelity, and positive semidefiniteness of the Gram matrix. No quantum advantage, hardware result or new learning-theoretic claim is made.

1 The construction

Qubit 0 is the first tensor factor and the CNOT control. For $x = (x_0, x_1) \in \mathbb{R}^2$,

$$U(x) = \mathrm{CNOT}\,(I \otimes R_Z(2x_0x_1))\,\mathrm{CNOT}\,(R_Z(2x_0)\otimes R_Z(2x_1))\,(H\otimes H), \qquad R_Z(\theta) = \mathrm{diag}(e^{-i\theta/2}, e^{i\theta/2}).$$

Since $\mathrm{CNOT}(I \otimes R_Z(\theta))\mathrm{CNOT} = e^{-i\theta Z\otimes Z/2}$, writing basis states by signs $s \in \{\pm 1\}^2$,

$$U(x)|00\rangle = \tfrac{1}{2}\sum_s e^{-i\varphi(s,x)}|s\rangle, \qquad \varphi(s,x) = x_0s_0 + x_1s_1 + x_0x_1s_0s_1,$$

so with $\Delta = x - x'$ and $\Delta_{01} = x_0x_1 - x_0'x_1'$,

$$K(x,x') = \left|\tfrac{1}{4}\sum_s e^{-i(\Delta_0 s_0 + \Delta_1 s_1 + \Delta_{01} s_0 s_1)}\right|^2.$$

This is one layer of the ZZ feature map of [1] with the simplified pair coefficient x_0x_1 (their map uses $(\pi - x_0)(\pi - x_1)$ and two layers). With one layer, and on two qubits, the kernel has the closed form above and is trivially classically computable; MK-1 is a teaching and testing artefact.

2 A defect in the original script

The original `mk1_moving_tower.py` (kept in `original/`) applied the middle $R_Z(2x_0x_1)$ to the *control* qubit. A diagonal gate on the control commutes with CNOT, so the two CNOTs cancel and no $Z\otimes Z$ phase is produced; the kernel becomes the product $\cos^2(\Delta_0 + \Delta_{01})\cos^2\Delta_1$. The package keeps this as `variant="legacy"` for reproducibility and uses the corrected map by default. Both facts ($\mathrm{CNOT}(D\otimes I)\mathrm{CNOT} = D\otimes I$ and the $Z\otimes Z$ phase for a target gate) are proved in Lean (`cnot_onControl_cnot`, `cnot_onTarget_cnot`).

3 What is machine-checked

`lean/MK1/FidelityKernel.lean` defines $K_\psi(x, y) = |\langle\psi(y), \psi(x)\rangle|^2$ for any family ψ in a complex inner product space and proves:

- `fidelity_self`, `fidelity_symm`, `fidelity_nonneg`, `fidelity_le_one`: $K(x, x) = 1$ for unit states, symmetry, $0 \le K \le 1$ (Cauchy–Schwarz);
- `norm_featureState`, `computeUncompute_eq_fidelity`: states $U(x)v$ from unitaries (linear isometry equivalences) are unit vectors, and $|\langle v, U(y)^{-1}U(x)v\rangle|^2 = K(x, y)$;
- `fidelity_gram_posSemidef`: for states in $\mathbb{C}^n$ and finitely many points, the matrix $[K(x_j, x_i)]$ is positive semidefinite, via $|\langle a, b\rangle|^2 = \langle a \otimes \bar{a}, b \otimes \bar{b}\rangle$ and Mathlib's Gram-matrix theorem.

All are `sorry`-free; `AxiomAudit.lean` shows only `propext`, `Classical.choice`, `Quot.sound`; CI rebuilds and audits on every push. These are standard facts; the Lean development checks them, it does not establish anything new. The Lean code does not model the specific gate angles beyond the CNOT identities; the agreement of the numerical circuit with the closed form is checked by the Python tests.

4 What is tested numerically

The Python tests (CI, every push) check for both variants: unitarity of $U(x)$; $K(x, x) = 1$; symmetry; $0 \le K \le 1$; compute–uncompute equals $|\langle\psi(x')|\psi(x)\rangle|^2$; agreement with the matrix-free closed form to 10^{-12}; Gram matrices symmetric with minimum eigenvalue $> -10^{-10}$; and the two CNOT identities.

5 Limitations

Exact statevector simulation, no shot noise or hardware; two qubits; no SVM is trained in the package (the DUAL/DECIDE stages are illustrative); no claim of quantum advantage. For the limits of such kernels see [3].

References

[1] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, J. M. Gambetta, Supervised learning with quantum-enhanced feature spaces, *Nature* 567 (2019) 209–212, doi:10.1038/s41586-019-0980-2.

[2] M. Schuld, N. Killoran, Quantum machine learning in feature Hilbert spaces, *Phys. Rev. Lett.* 122 (2019) 040504.

[3] J. M. Kübler, S. Buchholz, B. Schölkopf, The inductive bias of quantum kernels, *NeurIPS* 34 (2021).

Code: `https://github.com/BenFrohman/mk1-quantum-kernel`. License: Apache-2.0 (code), CC BY 4.0 (this note).